User Manual

Amrita Data-Center monitoring System

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Questions or Comments

If you have any questions for our technical support staff, please contact us at ammachilabs@amrita.edu

Introduction

Ammachi Labs is a multidisciplinary research Centre of Amrita University with a focus in technological innovation for social impact in the field of computer-human interaction, haptics, multimedia and virtual reality, with application areas in education, health care, defense and disaster preparedness.

Ammachi Labs extends Amrita University's mission of providing effective value-based education to include skill development at all levels and in all disciplines.

The core value of all research at AMMACHI Labs is to provide innovative technological solutions that are affordable to the masses.

Our Vision:

To strengthen the economic fibre and contribute to the social stability of the nation by empowering India's masses in a sustainable, cost-effective and scalable way, providing livelihood through innovative technological solutions and resources for education,healthcare, skill development and training.

Our Team:

Ammachi labs is an eclectic group of professionals in a wide range of disciplines. Our team consists of engineers (mechanical, electronics, electrical, computer science, production), machinists, 2D and 3D artists and animators, audio engineers, videographers and editors, instructional designers, teachers, UI/UX experts, social workers, psychologists, management and health-care professionals, artisans and master craftsmen, content writers and translators. Working together in a culture of human centered design, the Ammachi Labs team practices an integrated approach to development, from the research phase, all the way to deployment in the field.

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General Information

1. General Safety

Review the following safety precautions to avoid injury and prevent damage to this product or any products connected to it. To avoid potential hazards, fire or personal injury use this product only as specified.

- ❖ Only qualified personnel should perform service procedures.
- ❖ Take care to shutdown the System using System Shutdown command. Don't directly switch off the System using manual switch.
- ❖ Only if the System is OFF, then to restart the system manual ON/ Off switch can be used.
- ❖ Take care that UPS power is used to power the System.

2. Device Overview

Amrita Data Center Monitoring System is used to provide Safety for Servers in Server Rooms. It can monitor parameters like temperature, humidity and detect smoke in Server Rooms.

The Sensors can be wired to the Server Rack and help in monitoring the essential data. In case of the measured parameter crossing the set limits, it creates Alarm signal which is immediately shown in the website portal. It also sends Alert SMS messages to the Service personnel and the people, who are registered as Users in the Website portal. Also regular updates are sent via SMS to the registered Users.

2.1 Packing List:

The ADCMS Packing List consists of:

1. Server Monitor Device
2. Temperature/ Humidity Sensor with cable
3. Smoke Sensor with Cable
4. Power Cable

2.2 Device Specification

- Rated Voltage 220-240 V~

- Rated Frequency 50 Hz

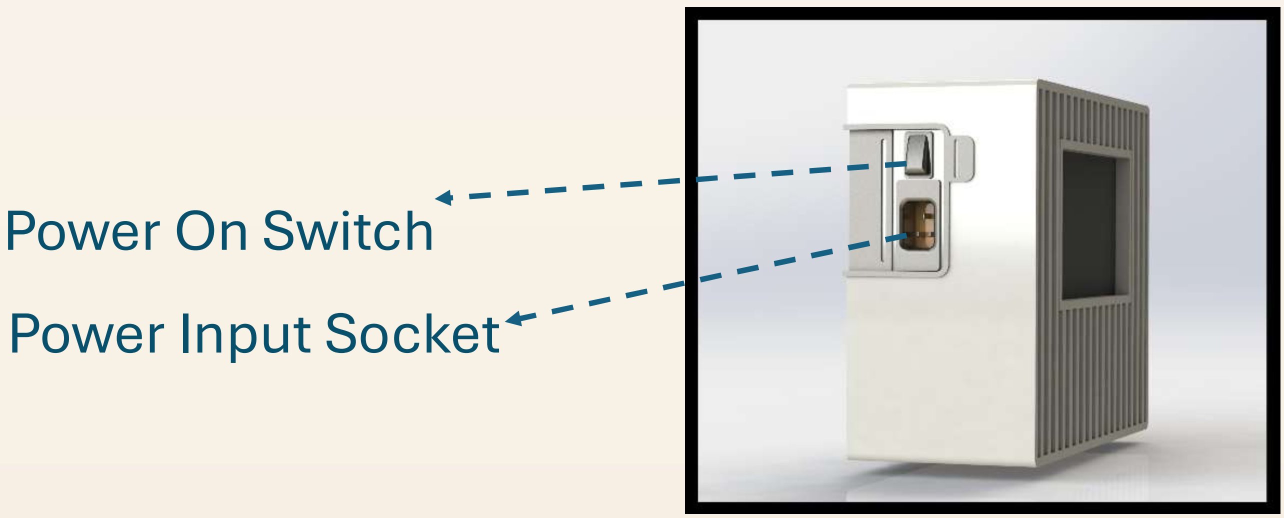
Getting Acquainted

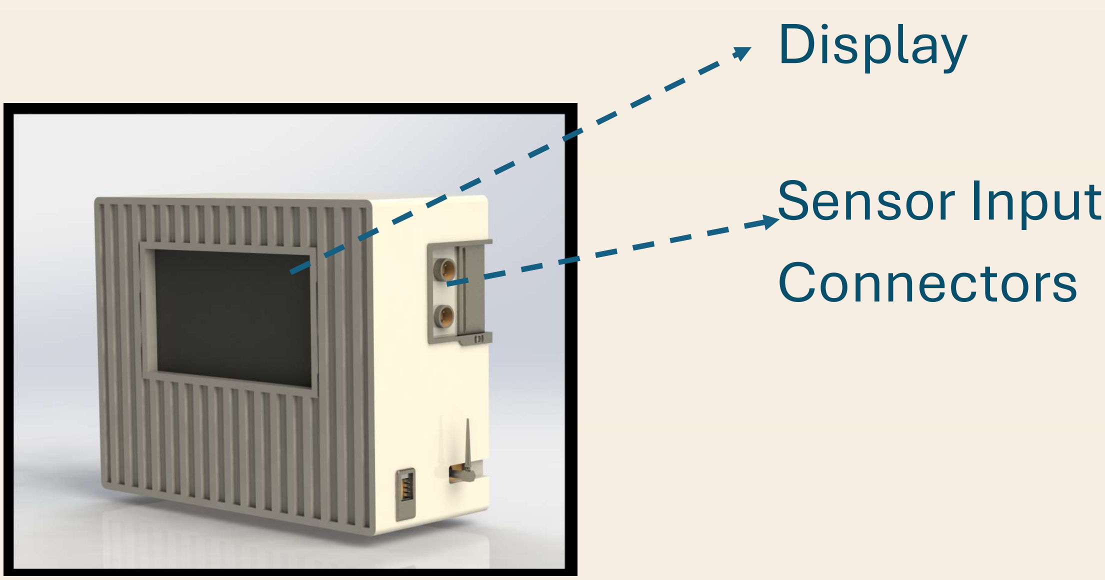
1. Familiarize the Parts of the System

The Amrita Data Center monitoring System has three parts to get familiarized

1. The Actual Device with Sensors
2. The Website Portal
3. Tablet on the Entrance, which displays the website portal. (Optional)

2. Physical Device





SIM Socket

3. Parts of the System:

❑ Display

Display is the front-end interface which shows the temperature, humidity and smoke detection Data and allows the Users to interact with the System.

❑ Sensor Input Connectors

Sensor input Connectors help to connect the sensor cable to the system.

❑ Power On Switch

Power On Switch is used to power on the System. The System is also battery powered. Battery is internal to the System and the System needs to be connected to the 230V power for charging the battery. Incase of power failure, the system can work on battery for more than half hour. The User should properly shut down the System, if the power failure prolongs more than half hour.

> ***NOTE:*** *As the System is a mini computer, the System need to be properly shutdown by clicking shutdown command on the display or web portal.*

❑ SIM Socket

SIM Socket allows one to insert the SIM.

4. Installation Procedure:

The System will be installed by the Service Personnel. It has to be connected to nearest the UPS Power Outlet. The Temperature Sensor is kept near the Server rack to sense proper temperature and the Smoke Sensor on the inside top of the rack to sense the smoke

Website Portal

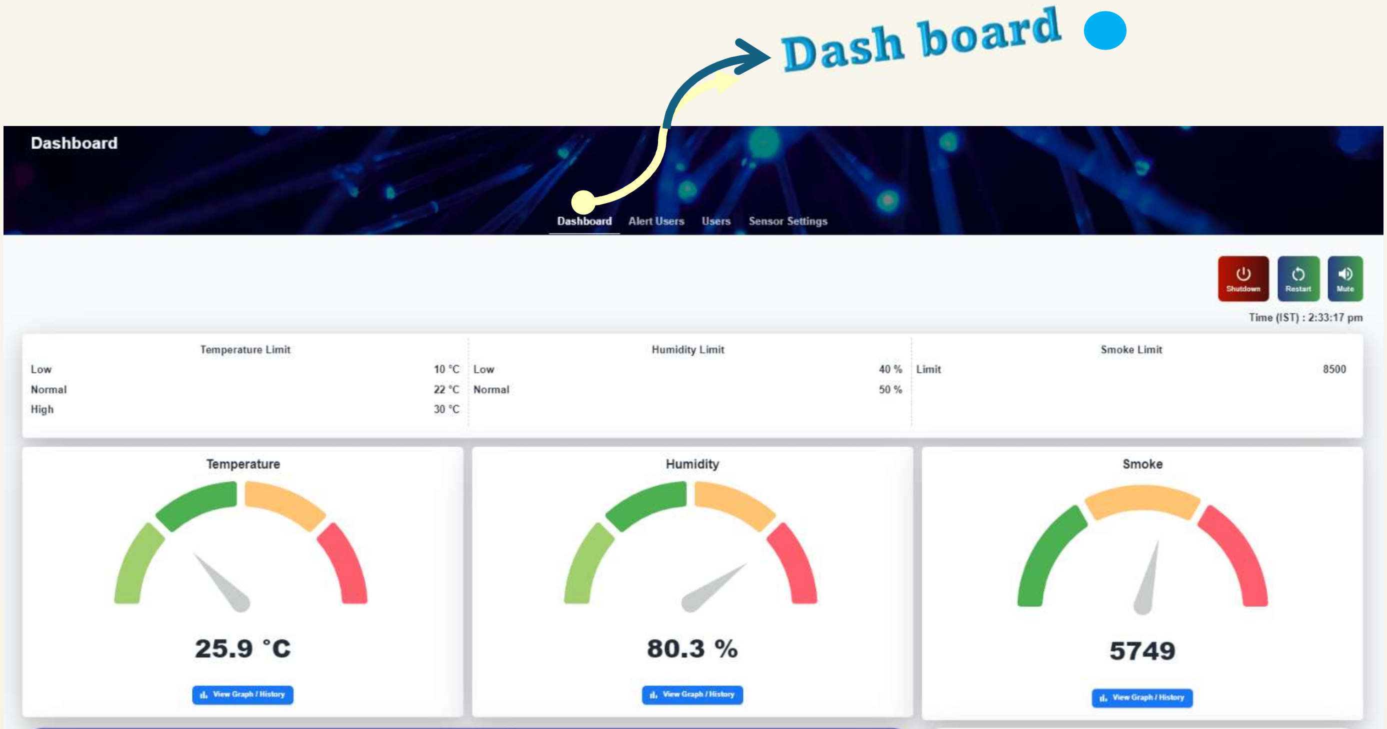
The Website portal is the main output portal for the Server room personnel to monitor the Servers.

The Pages of the website portal are (when logged in as Admin User only)

- Dashboard
- Alert users
- Users
- Sensor Settings

They are explained below. The First page is Dashboard.

Dashboard

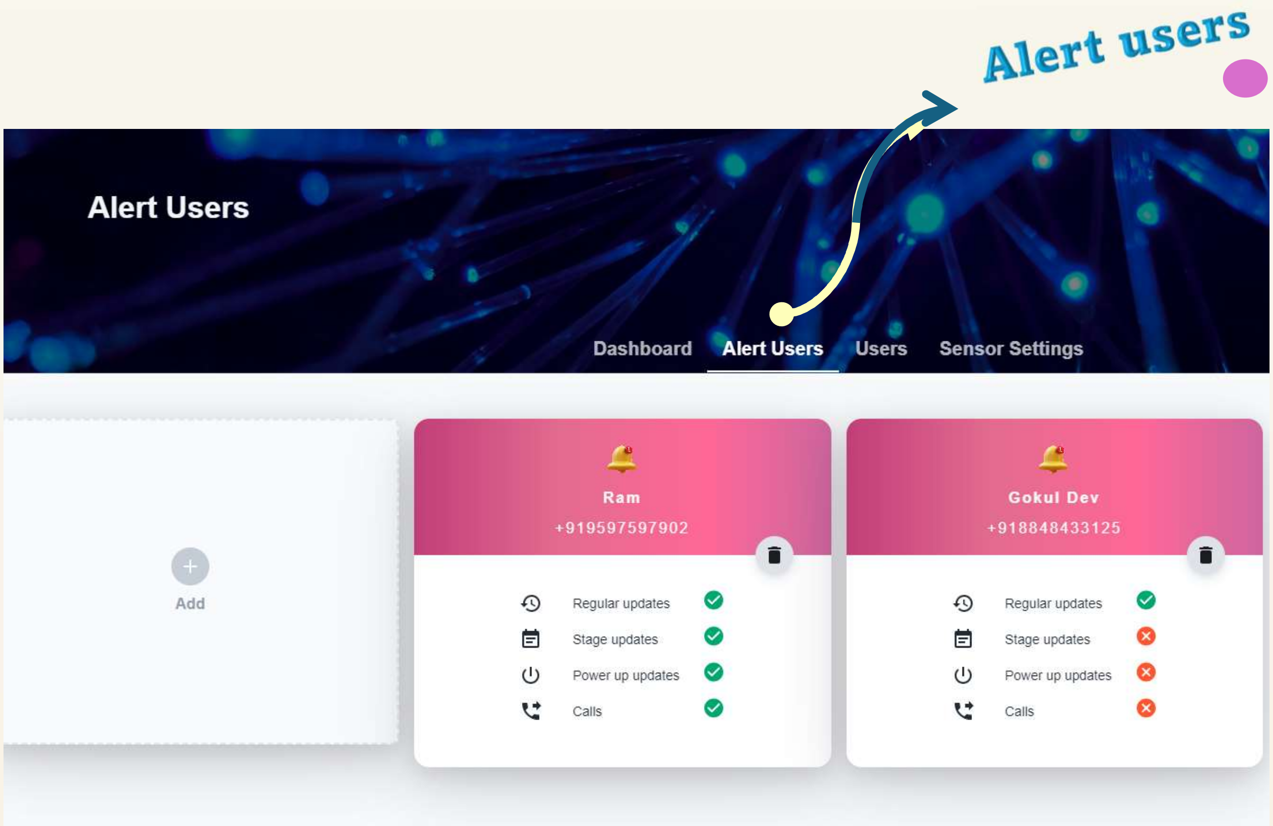


Dashboard

Dashboard is the first page displayed.

- In this **page**
  - The sensed data of temperature, humidity and smoke are displayed .
- The other options avialble in this page are :
  - **Shutdown** option : The Shutdown option allows one to shutdown the system
  - **Restart** option : The Restart option allows one to restart the system
  - **Mute** Option : It allows one mute the Alarm

Alert Users

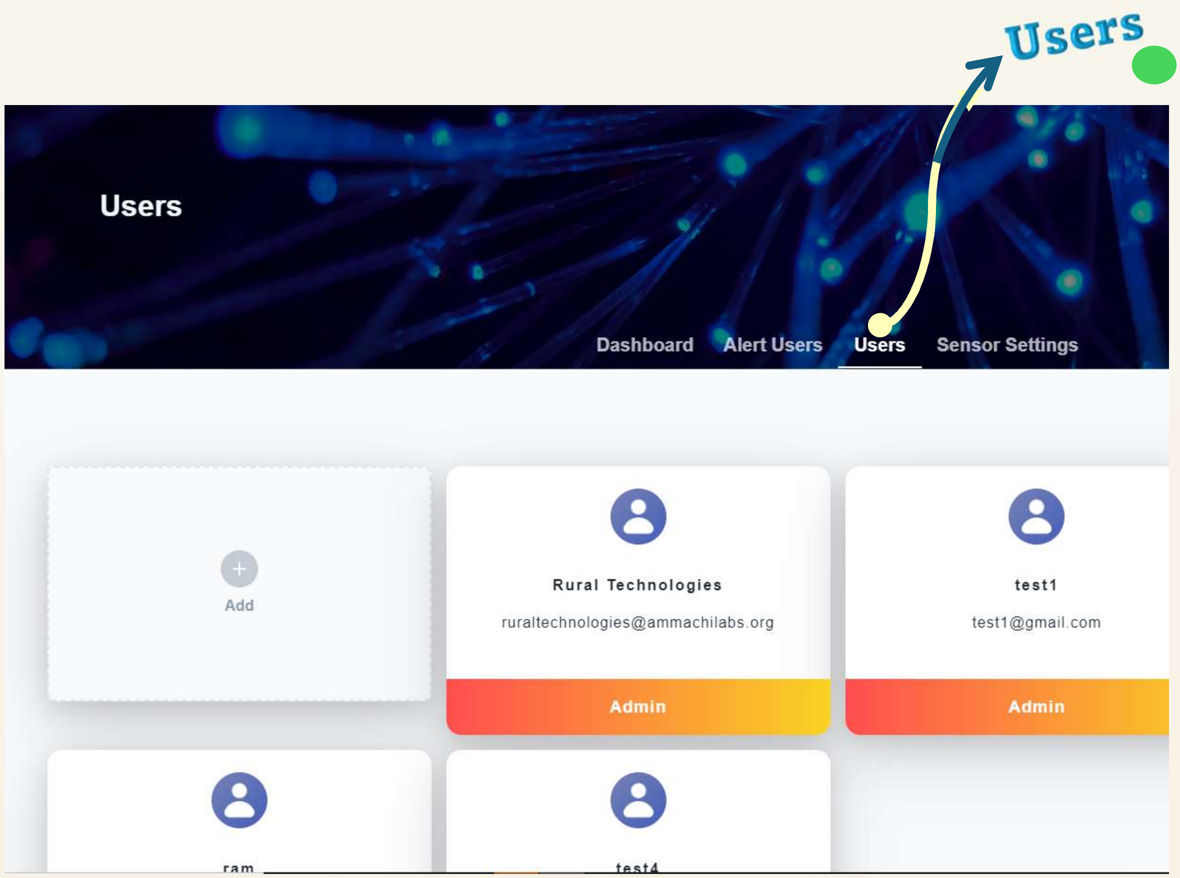


Alert Users

Alert Users is the next page.

- In this **page**
  - The phone numbers of Users to whom the data need to be sent can be added. Also there are options to select the type of data to be sent .
- **The Types of data** are
  - Regular Updates – regular data is sent to the User on hourly basis.
  - Stage Updates – data is sent only when there is an alarm
  - Power Updates – whenever the System is Shutdown and restarted the alert is sent
  - Calls – When there is an alarm a call is made to the User

Users



Users

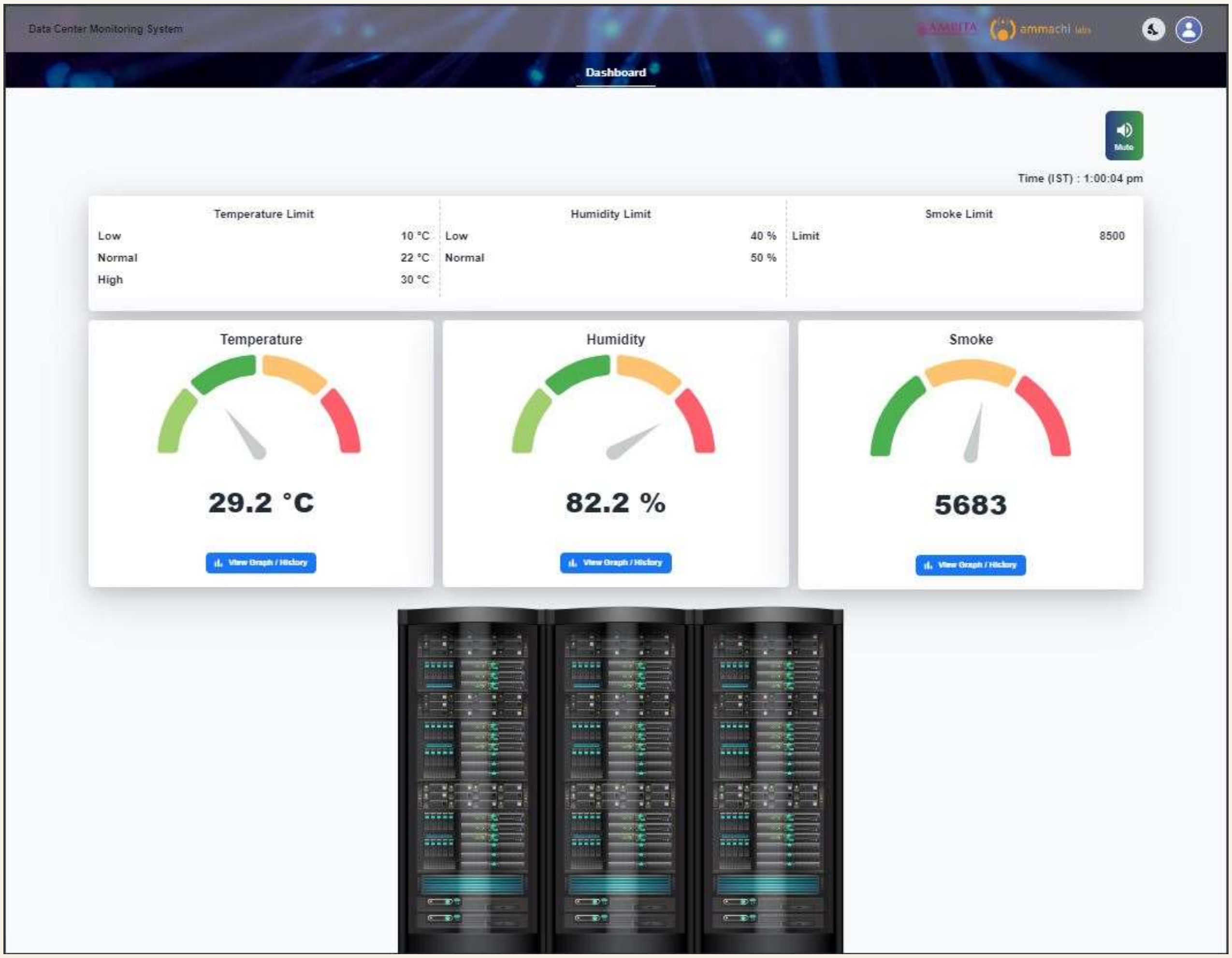
Users is the next page.

- In this page the Users who can access the web page can be added. There are two types of Users
    - **Admin Users**
    - **Users**

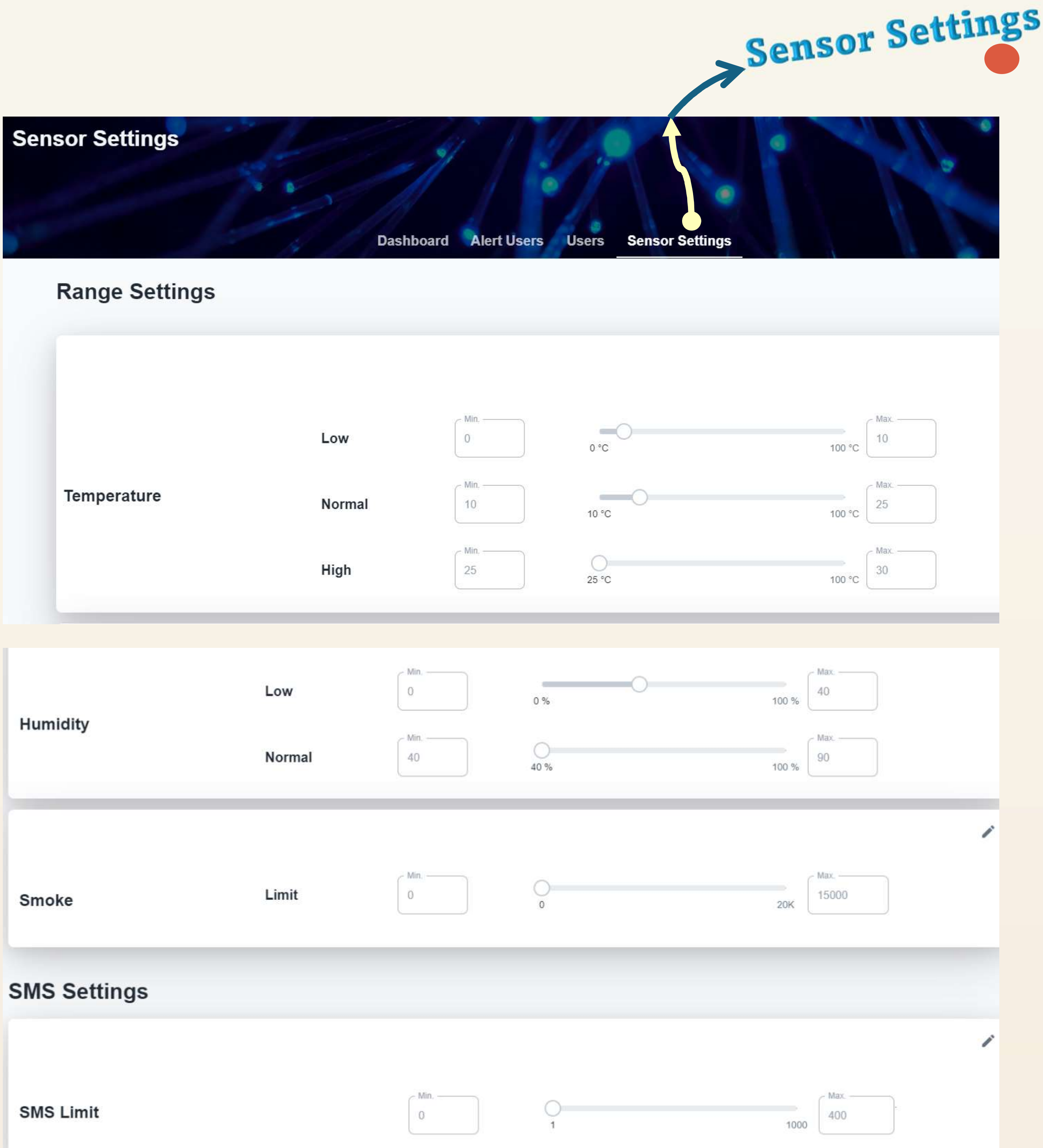
- **Admin Users** can have full access and edit all the options as shown above.

- **Users**

  When **logged in as only Users –** one can only view the displayed data and can mute the Alarm. The display looks like this in User view as shown below.



Sensor Settings



Sensor Settings

This is the next page.

- In this Page,
  - The Admin User can set the **range of** Temperature, Humidity and Smoke parameters. Here, the System need not be rebooted after the ranges are adjusted.
  - In Temperature Limit Setting, there are 3 types of Ranges
    - Low
    - Normal
    - High

    This is to show in which category the sensed temperature falls.

    The Admin User can set the lower and higher Limits of each of the ranges as shown in the picture above.

- In the Humidity Limit Setting there are 2 types of Ranges
  - Low
  - Normal

  This is to show in which category the sensed humidity falls.

  The Admin User can set the lower and higher Limits of each of the ranges as shown in the picture above.

- In the Smoke Limit Setting the lower and higher limits can be set.

- In the SMS Setting, the number of SMS that can be sent per month can be set.

Note: The System need to be rebooted after SMS Limit is changed or if a User is added.

System Offline

Due to any reason if the System is Switched off and in that state, if we try to login, the website portal shows **“System Status as OFFLINE”** as shown below.